

Alset IoT

“Hug The Lanes”



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The future is to use IoT to make cars safer, less costly, and more
efficient!

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Section 1:

Introduction

The 21st century has seen exponential growth in technological advancements, with the past 23 years celebrating achievements from the earliest smartphone models of contemporary tech giants to machine-learning techniques used in AI training. The overarching goal appears to be streamlining human life by reducing manual effort. For instance, smart home devices can turn on lights with a voice command, tablets and touchscreen laptops offer endless writing space, and self-driving vehicles replace both manual and automatic driving.

The future in store for self-driving cars can be greatly beneficial for modern-day and prospective drivers. More accidents can be prevented with the help of a car's algorithm that is aware of all its surroundings all the time. Those who have trouble driving or need accommodations to do so may have an easier time traveling in their own vehicles. Of course, a self-driving vehicle does not totally replace a human driver who must still take responsibility when driving. Still, as we have seen with other emerging technologies, it could make transport easier.

Using the Unified Process method, our team hopes to manufacture high-level software that can simulate that of a self-driving car. We will require both functional and non-functional components to construct our algorithms. Some features that we aim to include are: autonomous navigation and control, obstacle detection and avoidance, real-time traffic analysis and route planning and more. For the first release, for example, we plan to offer a first prototype of sensor detection. This will alert the motor vehicle about other objects in close proximity to the point of a potential collision or danger. After this first release, revisions will be performed and edits will be made to improve upon our work. Then, subsequent releases will build upon their prior ones and follow suit.

This project is one we take seriously and maturely because there is no room for error. When developing software for items as potentially dangerous as a self-driving automobile, any incorrect bit of code can have fatal consequences. Therefore, the performance and reliability of our final product must most accurately reflect our dedication to this project. Fortunately, we can most sensibly accomplish this by following a well-structured model for developing our software. We aim to adhere to the Unified Process as our role model. In this way, we can make initial plans, gather necessary materials and resources, revise our plans if necessary, and then continue to execute them in a well controlled environment. This will also prevent us from forgetting to implement certain features or implementing ones that we eventually decide against.

This team's members have had the opportunity to gain experience in different departments and subsets of computer science and cybersecurity. Our computational skills range from basic operations on simple, linear data structures to a deeper understanding of computational theory and its representations. Furthermore, we collectively possess an arsenal of skills and qualities, including being task-oriented, patient and analytical, that will help us persevere in our project. In all, we are prepared to take on the challenge of developing this software, and we will not be satisfied until the final product best reflects our tenacity.

Additional Information

- 1. Goal: Developing a systematic infrastructure using software and hardware to create a car with autonomous navigation and control**
- 2. Datasets**
 - a. Hardware**
 - i. GPS data (positioning and navigation)
 - ii. Camera data (detecting road signs, traffic lights, and obstacles)
 - iii. Ultrasonic sensors (measuring distance from nearby objects)
 - iv. Radars for (detecting other vehicles and objects in the road)
 - b. Software**
 - i. Information from the driver's control inputs, such as steering wheel and pedals.
- 3. Use IoT to make cars safer, less costly, and more efficient**
 - a. Use IoT to make decisions or provide information to the driver
 - b. Use IoT edge devices that can capture data from the car and environment
 - c. Use an analytic engine (IoT Engine) on board to process information and make decision that is passed on to the Car Control System (CCS)
 - d. Provide capability for the driver to enter commands and receive status
 - e. Download the latest rules and updates for operation from the Fog/Cloud into the IoT Engine
 - f. Upload the car data to the company cloud

Team Roles and Qualifications:

Leader: Anne-Marie Akoh

Members: Joey Bonetti, Allan Joseph, Octavio Morales

<u>Name: Anne-Marie Akoh</u> <u>CS Courses Taken:</u> CS 115, CS 146, CS 284, CS 135, CS 382, CS 385, CS 334, CS 306 <u>Languages:</u> Python, Java, Javascript, HTML, CSS, C++, Assembly <u>Qualifications:</u> Writing, Patience, Task-Oriented	<u>Name: Joey Bonetti</u> <u>CS Courses Taken:</u> CS 115, CS 146, CS 284, CS 135, CS 382, CS 385, CS 306 <u>Languages:</u> Python, Java, Javascript, HTML, CSS, C++, Assembly <u>Qualifications:</u> Analysis, Curiosity, Task-Oriented
<u>Name: Allan Joseph</u> <u>CS Courses Taken:</u> CS 115, CS 101, CS 284, CS 135, CS 382, CS 385 <u>Languages:</u> Python, Java, Assembly, Swift, C, C++ <u>Qualifications:</u> Analysis, Curiosity, Coding	<u>Name: Octavio Morales</u> <u>CS Courses Taken:</u> CS 115, CS 101, CS 284, CS 135, CS 382, CS 385 <u>Languages:</u> C#, Python, Java, C, C++, Assembly <u>Qualifications:</u> Coding, Outgoing, Patience

Section 2:

Functional and System Architecture:

We shall now detail the functionality of the architecture we plan to implement for our self-driving car. It is incredibly important that the architectural interfaces we develop connect seamlessly with one another in order to ensure a fluid and unbroken highway of data transmission and communication between components. Importantly, we look to involve the following sensory modules; radio wave detectors, sonar detectors, three-dimensional scanning cameras, and GPS navigation features. Ideally, we need to ensure that each part works flawlessly for the self-driving car to be used publicly. As shown in *Figure 2.1*, a part will more than likely have a direct impact on the performance of hardware used later, so the failure of one part can result in the failure of the intentional self-driving car functionalities. When the architecture is implemented, it will result in many of the features for the self-driving car.

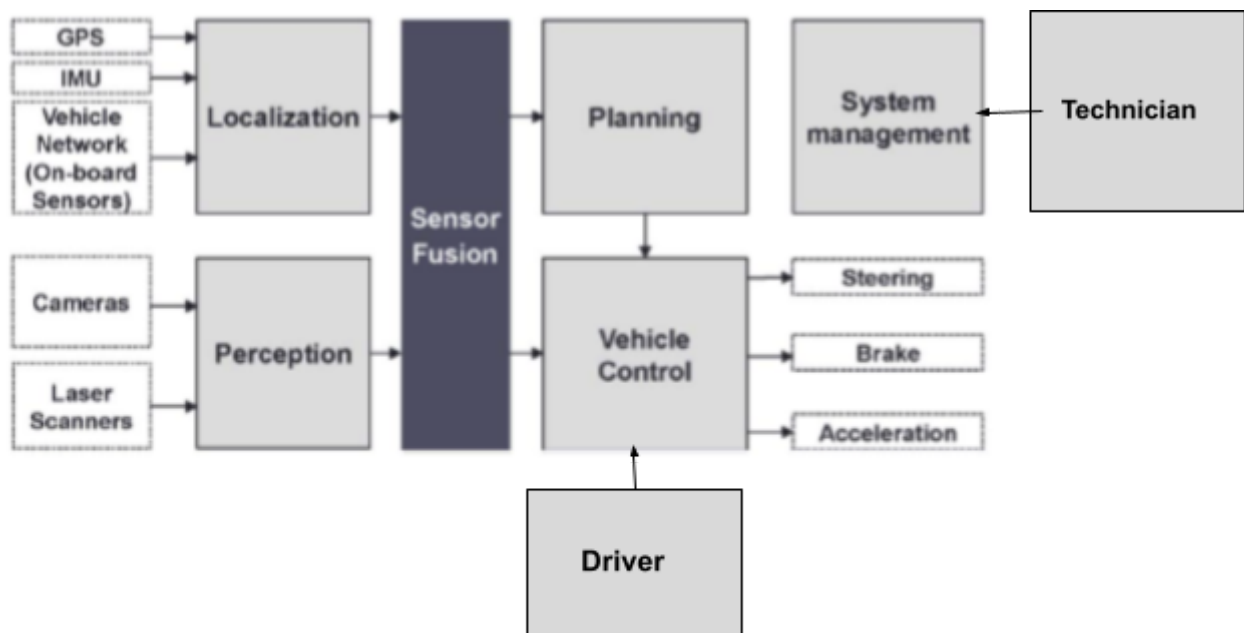


Figure 2.1

Linked Components:

2.1 Sensor Fusion

The cooperation and collaboration of different sensors embedded in the car's architecture will equip the vehicle with enough information to make informed and intelligent decisions about its movement. This sensor fusion is necessarily made up of both localization and perception, each with their own subsets of components to build them. We analyze each individually.

Localization:

Sensors can be used to detect and analyze the environment around the vehicle and provide real-time feedback to the driver. For example, cameras and lidar sensors can detect other vehicles, pedestrians, and obstacles on the road, while GPS sensors can provide location information. By integrating these sensors into the driver assistance system, the architecture can be validated for its ability to provide accurate and reliable information to the driver, thus ensuring their safety.

Perception:

To ensure that the moving vehicle has a good and accurate understanding of its current position and surroundings at any point, we will rely on the information relayed by perception-based sensors. Both cameras and laser scanners will receive environmental input from around the car, convert it into information readable by the vehicle, and finally send that data to the other components of the architecture. This way, decisions of movement that require a current understanding of the car's position can be well judged and safely made.

2.2 Planning and Vehicle Control

The critical information received from the sensor fusion component in this vehicle's architecture will be necessary in planning and controlling movement of the car. The combined data provided by both the localization and perceptions fields of sensors is required in order for the self-driving car to decide where to move next, when it should do so, and at which speed to do so. These planned decisions will feed into the vehicle control component which will physically carry out those plans. For example, the degree by which to turn the steering wheel, the speed at which to hold down the brake pedal, and the duration for which the car should be accelerating are all possible factors that may be directed by controlling the vehicle.

However, despite the car possessing self-driving capabilities, it is still necessary to allow for user input - that is driver input. The driver of the car must also be able to influence vehicle control. This acts as a safety measure for when sensory information is either miscommunicated in transmission or incorrectly received in the first place. Similarly, there may be times when the driver makes unintended or potentially dangerous decisions. In which case, the correctly received sensory input will prevent the driver from carrying out those decisions. The two keep each other in check in this way.

2.3 System Management

In self-driving vehicles, sensors play a critical role in making decisions and taking actions. For instance, radar sensors can detect the distance and speed of surrounding vehicles, while ultrasonic sensors can detect the proximity of obstacles. By using machine learning algorithms to process the data from these sensors, the self-driving system can make informed decisions and take necessary actions to ensure safety. Any changes made to the architecture must take into account the impact on the accuracy and reliability of the sensors and their ability to make decisions. Therefore, it is important to trust that the system of the vehicle is always as up to

date as possible. A qualified and professional technician will be the recipient of the current status of the car's performance. Should there be any issues or potential hazards, the technician will revise the car's architectural components, identify the issues, and reconstruct faulty parts if necessary. Doing this regularly will ensure that the self-driving vehicle is well maintained and constantly safe to travel in.

System Admin

In a system administration setting, sensors can be used to monitor the health and performance of the network and cloud infrastructure. For example, temperature and humidity sensors can monitor the server room, while network sensors can monitor network traffic and identify potential security threats. By using this data to analyze the performance of the network and cloud connection, system administrators can run diagnostic evaluations to validate the architecture, detect issues or potential crises, and make informed decisions about making any drastic changes.

(info to be used in section 3 instead...)

Radio Wave Detectors

The inclusion of sensory detectors of radio waves on the electromagnetic spectrum will be useful in providing information about an object's position in relation to items around it. In terms of a self-driving car, these sensory detectors can pick up radio wavelengths of varying frequencies depending on the proximity of a surrounding object. The closer the object is to the car, the more wary our sensors will be until it reaches a point of possible collision. The detectors will track the distance between the car and other nearby objects, record the angles of closeness between them and measure, if applicable, the speed at which they are traveling. In this way, necessary information will be provided to the car's other functional components so that they can further build off it.

Sonar Detectors

As important as it is to have radio wave detection of physical objects, it is to have sonic detection of our surroundings. With the help of the sonar functional capabilities we shall implement, our audible detecting devices will be able to determine the approximate closeness of an based on how soon sound waves are transmitted back. Emitting sound waves from the architectural system and tracking how long it takes to receive their echoing counterparts back will work hand-in-hand with our radio wave detectors. These two sensory subsets will provide a combined effort of judging the position of the car in relation to potential hazards around it. To ensure an extra safety check, these sonar detectors will give and receive information to and from the radio wave detectors. Readings of the same external object's proximity should be the same if not minutely different between the two. When this is the case, we will have ensured that our car has reliable means for understanding its own positional status.

Three-Dimensional Scanning Cameras

While intangible methods for determining current position are incredibly useful and important, they are not necessarily user-friendly. This is essentially because a passenger may be able to derive little to nothing from radio wave and sound wave detectors alone. Humans tend to be more visual creatures and can learn a great deal simply from what they see. This is why it is important for our car to also include cameras to provide a sense of direction, position, and proximity to other objects. Our three-dimensional scanning cameras will receive information provided from both the radio wave and sonar sensors and incorporate into projecting a visual image that represents the numerical data. In this way, not only passengers but the car itself will have a visual representation of where they currently are, where they are going, and what is surrounding them. It is also possible that, in rare cases, radio wave or sonar sensory systems may be faulty. In this event, having image sensors can still inform the car of where it is in relation to others. However, these cameras can only collect visual data close enough to be in camera view. This is why it is necessary to have varied means of positional detection to account for things that may not be visually detectable. Therefore, the radio wave sensors, sonar sensors and these visual image sensors will communicate with each other to gather, read, and interpret information about the car's position. Should one functional component fail, there will still be others to make up for it as much as possible until the necessary repairs are made.

GPS Navigation Features

Of course, it is necessary that a self-driving car has sensory architecture to relate its position to that of other objects or items nearby. Still, this does not and cannot act as any sort of geographical locating substitute. Essentially, the aforementioned sensors can describe where a car is in a local and relative sense, but it cannot do so in a global sense. These sensors do not know where in the world they are but do know where they are relative to objects just close enough to be detected. This is why it is important for us to also incorporate a geolocating component in our car's functional architecture. This way, our self-driving automobiles can have an understanding of where they are in the world. With the use of nationwide and worldwide satellite systems of navigation - more commonly known as Global Positioning System, or GPS - our car's geolocating sensors can read information provided by them and share it with the car's other functional components. As such, the radio wave, sonar and image-detecting sensors will interact with each other to sense the local position of the car while these geographical ones will interpret global position. This will be a great aid in directing the car to travel to certain locations using a navigation system. Not only will the car drive with full comprehension of how close it is to other objects, but it will also be able to travel to specific geographic coordinates based on the GPS satellite information it receives on the way there.